VeriReason:
Intervention-Based Claim Verification and Repair for LLM-Generated Credit Denial Explanations

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Research Idea Proposal
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Section 1

Motivation

Why Claim-Level Verification?

- **Requirement.** A denial must state the *specific, principal reasons* (Regulation B, 12 CFR 1002.9 and its Official Interpretation).
- **Practice.** Reasons come from SHAP and are now rewritten into plain language by LLMs.
- **Problem.** Studies report that LLMs name the influential factors but are less reliable on *direction*; a fluent, wrong reason is a compliance risk.
- **Gap.** Evaluations score the whole text, not which sentence is wrong or how to fix it.
- **Need.** Check every claim against the model; repair only the claims that fail.

Regulation B and its Official Interpretation remain in force; CFPB Circular 2022-03 was withdrawn on 12 May 2025 and is background only.

Objectives and Scope

Goal

A verification-and-repair framework that checks each claim of an LLM-generated credit denial explanation against the behavior of the scoring model, identifies the type of each error, and selectively repairs only the claims that fail.

Objectives

1. Define an error taxonomy grounded in the principal-reasons requirement.
2. Decompose LLM explanations into typed, checkable claims.
3. Verify each claim against the scoring model through dependency-aware interventions.
4. Repair only the claims that fail, preserving supported content.

Scope

- Tabular credit scoring: gradient-boosted trees (primary), logistic regression (validation)
- English explanations for denied applicants, from several LLMs and prompting setups
- Public credit datasets; TreeSHAP attributions
- Not covered: legal-compliance certification, fairness auditing, multimodal models, explanation drift

Research Questions

RQ1 Error Characterization

How often, and in which ways (factor, direction, magnitude, omission, specificity), do LLM-generated credit denial explanations contradict the behavior of the scoring model?

RQ2 Verification

Does dependency-aware, intervention-based verification detect these errors more reliably than attribution-sign checks, LLM judges, and general claim-verification frameworks?

RQ3 Repair

Can selective claim-level repair reduce errors while preserving supported claims, compared with full regeneration and claim filtering?

Section 2

Problem Statement

Setting and Notation

- Scoring model $f : \mathcal{X} \to [0, 1]$ (predicted default risk); applicant x is denied when $f(x) \geq \tau$.
- Features are partitioned into **reason groups** $G = \{g_1, \ldots, g_K\}$: payment history, credit utilization, recent inquiries, income and affordability.
- $\varphi(x)$ is the TreeSHAP attribution of f at x. An LLM writes an explanation e from $(x, f(x), \varphi(x))$; some setups omit $\varphi(x)$.
- An extractor splits e into claims $C = \{c_1, \ldots, c_m\}$, each $c_j = (g_j, \delta_j, \mu_j)$ with its text span:
  - $\delta_j \in \{\text{raises risk}, \text{lowers risk}, \text{neutral}\}$ (stated direction)
  - $\mu_j \in \{\text{principal}, \text{contributing}, \text{minor}\}$ (stated prominence)

Reference Effect

$$\Delta_g(x) = f(x) - \mathbb{E}_{\tilde{x}_g \sim Q_g(\cdot \mid x_{-g})}\Big[f\big(h(\tilde{x}_g, x_{-g})\big)\Big]$$

- x_{-g}: the applicant's features outside g; $\tilde{x}_g$: a plausible alternative value of group g.
- Q_g: conditional (on-manifold) sampler on a reference population; h recomputes derived features (e.g., debt-to-income).
- Verified direction $\widehat{\delta}_g = \mathrm{sign}_\varepsilon\big(\Delta_g(x)\big) \in \{+, 0, -\}$: $\Delta_g(x) > \varepsilon$ means the applicant's values in g *raise* predicted risk.
- Risk-increasing groups $G^+ = \{g : \Delta_g(x) > \varepsilon\}$, ranked by $\Delta_g(x)$; principal reasons are the top k of G^+ (Reg. B commentary: usually $k \leq 4$).

A defined reference, not ground truth

$\Delta_g(x)$ depends on Q_g and the reference population: verdicts are reported under several references.

Error Taxonomy

ID	Error	Meaning	Example	Check
E1	Unsupported factor	Described as contributing, but no meaningful effect, or not a model input	"Your age counted against you" (age not used)	$\delta_j \neq$ neutral but $\widehat{\delta}_g = 0$, or g not an input
E2	Wrong direction	Stated direction opposite to model behavior	"No prior loans raised your risk" (model: neutral or favorable)	$\widehat{\delta}_g \neq 0$ and $\widehat{\delta}_g \neq \delta_j$
E3	Wrong prominence	Minor factor shown as principal, or the reverse	Small inquiry effect listed as the main reason	rank of g in G^+ vs. μ_j
E4	Omission	A verified principal group is missing	Top driver (utilization) never mentioned	$\mathrm{Top}_k(G^+)$ vs. mentioned groups
E5	Non-specific	Cannot be mapped to a group, or too generic	"Your overall credit profile was weak"	flagged non-checkable
E6	Invalid recourse (optional)	Action would not change the decision, or targets an immutable attribute	"Increase your income by 2%" (no flip)	$f(h(x')) \geq \tau$ for the suggested x', or immutable g

x': applicant profile after the suggested action. Ranking within G^+ uses the signed $\Delta_g(x)$, so a favorable group is never a reason for denial.

Section 3

Related Work

LLM Explanations in Credit and Finance

- **Zandi et al. (2026):** on Freddie Mac data, LLM narratives name influential factors reliably but are less reliable on direction.
- **Geng et al. (2026):** LLMs work best as narrative interfaces, not substitutes for formal attribution methods.
- **AIMarri et al. (2025):** when an LLM explains its own default predictions, its feature-impact statements often disagree with SHAP on financial tabular tasks (the LLM is the classifier there, unlike our setting).
- **LEXMA (2026):** RL fine-tuning for mortgage decisions; experts and consumers preferred it to the base model.
- **Principal reasons:** deployers take top attribution features; one lender groups SHAP values and adds a directionality score.

Takeaway

LLM credit explanations can be wrong, yet none are verified by intervention or repaired claim by claim.

Faithfulness and Claim Verification

- **FActScore, RARR:** atomic-fact scoring and evidence-grounded revision; not faithfulness to a model's behavior.
- **Counterfactual tests:** Atanasova et al. (2023), Siegel et al. (75 models, 13 families), Walk the Talk (2025); Faithful Serum (2026) adds attribution-guided steering.
- **FAX (2026):** decomposes drafts into claims and tests them with tools that query the target model; refuted claims are removed or rewritten, inconclusive ones are not presented as verified. Evaluated mainly outside credit (main benchmark: an RL environment); German Credit is one of three TalkToModel tabular sets, and the authors caution that such benchmarks can conflate task accuracy with model-specific faithfulness. *Closest prior work.*
- **EDCT** (vision-language models), **TalkToModel** (conversational tabular explanation).

Takeaway

FAX is a main baseline. It has no notion of principal reasons, omission or prominence, and no constraint to preserve supported claims. Missing: a credit-specific taxonomy, interventions for correlated features, and repair that preserves supported claims; the pilot tests whether FAX suffices.

Positioning

Capability	LLM-in-credit	General faithfulness (CT/CCT, Serum)	FAX	Ours
Credit-specific setting	Yes	No	Partial (German Credit only; mainly non-credit)	Yes
Claim-level decomposition	n/r	Partial	Yes	Yes
Intervention-based verification	n/r	Yes	Yes (faithful tools)	Yes
Credit error taxonomy	Partial (direction)	n/r	n/r	Yes
Correlated-feature interventions	n/r	n/r	n/r	Yes
Selective repair	n/r	Partial (steering)	Partial (remove or rewrite; no preservation constraint)	Yes

n/r = not reported in an abstract-level reading, not proven absent; to be verified in full texts (Week 0).

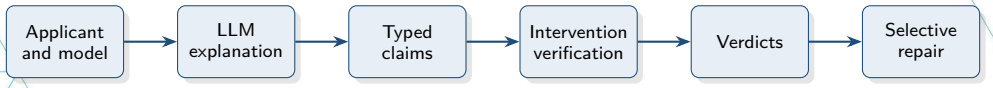
Section 4

Proposed Method

Method Overview and Claim Extraction

Core Question

Which claims contradict the model's behavior, what kind of error is each, and how can the explanation be repaired without changing supported claims?



- An LLM extractor with a fixed schema returns group, direction, prominence, optional action, and the character span.
- Spans are preserved so repair is local; unmappable claims are labeled E5, not discarded.
- Extractor quality is measured on a manually annotated subset and carried into the ablations.

Dependency-Aware Verification

- **Group-level replacement:** related features move together; derived features are recomputed by h.
- **Plausible alternatives:** Q_g is a conditional sampler (nearest neighbors in x_{-g}, or a conditional generative model), not independent marginals that create impossible profiles.
- **Claim verdicts:** supported, or E1, E2, E3, E5 (E6 for recourse claims), using $\widehat{\delta}_g$, the rank of g in G^+, and the flip test.
- **Response level:** omission (E4) and consistency of principal reasons across near-identical applicants.
- **Reference sensitivity:** conditional sampler, population median, approved-applicant reference; a SHAP-sign verifier runs on the same claims.
- Each verdict is stored with its evidence ($\Delta_g(x)$, rank, samples): an **audit record**.

Selective Repair

1. Build a fact sheet per flagged claim (group, verified direction, rank).
2. Regenerate only the offending span, or insert a sentence for an omission.
3. Re-verify; after a few attempts fall back to a templated sentence.
4. Text diff: supported claims must stay verbatim.

Before

✓ Payment history
✗ Inquiries (E3)
✓ Utilization

⇓

After

✓ Payment history
✓ Inquiries
✓ Utilization

Preservation Objective

Supported claims remain unchanged; the number of edited spans equals the number of verified errors.

Validating the Verifier

- **Logistic regression** $f(x) = \sigma(w^\top x + b)$: the group effect in logit space, $w_g^\top \left(x_g - \mathbb{E}_R[x_g]\right)$, is analytic and gives the true direction and ranking. We measure how often the verifier agrees with it.
- **Semi-synthetic credit model:** planted rules give known reasons per applicant, testing the verifier under correlated features and interactions.
- **Expert annotation:** a small set of XGBoost explanations labeled by annotators with credit or finance background.

Note: with a conditional sampler and the sigmoid, $\Delta_g(x)$ itself is not closed-form; it is estimated by sampling.

Section 5

Experimental Plan

Datasets, Models and Generators

Dataset	Role	Note
Home Credit Default Risk	Primary (pilot)	Linked tables; Kaggle account and acceptance of the competition rules required; do not publish raw data
German Credit (UCI, id 144)	Comparability	CC BY 4.0; used by prior work and as one of FAX's three tabular sets; added in Weeks 3–5
FICO HELOC	Comparability (optional)	Request access via the FICO form; special negative codes need handling
Lending Club / Freddie Mac	Secondary replication	Lending Club copy on Kaggle is user-uploaded: check its licence, remove leaking columns; Freddie Mac needs registration and cannot be redistributed

- **Models:** XGBoost (primary; random forest if time permits), logistic regression (validation), TreeSHAP.
- **Caveat:** by default, TreeSHAP for XGBoost attributes the log-odds margin; these values add up in logit space, not probability space, so turning them into probability-scale statements is itself a possible source of error.
- **Generators:** 3–4 LLMs (small and large open-weight, one proprietary) $\times$ 3 prompts (features; +SHAP; +SHAP and group dictionary), several samples per applicant.

Baselines

ID	Baseline	What it tests
B0	Direct LLM explanation	Raw error rates without any verification
B1	Deterministic reason-code template from SHAP groups	Industry style: near-zero error, limited readability
B2	SHAP-sign verifier	Whether attribution signs suffice to check claims
B3	LLM-as-judge (with / without model access)	Whether a judge LLM detects direction and prominence errors
B4	FAX-style verify-and-filter (re-implemented for credit)	Closest general framework; central to the go/no-go decision
B5	RARR-style revision	Generic evidence-grounded selective revision
B6	Full regeneration with verified facts	Repair without selectivity; measures preservation cost

Evaluation and Ablations

Errors and verification (RQ1, RQ2)

- Claim-level error rate by type (E1–E6); share of explanations with ≥ 1 error
- Precision, recall, F1 per type vs. the reference (analytic on LR; expert labels on an XGBoost subset)
- SHAP-sign vs. intervention disagreement, split by feature correlation

Repair and cost (RQ3)

- Residual error; supported claims unchanged; edit distance
- Principal-reason precision / recall vs. verified top-k; omission; stability
- Readability (LLM judge plus small human check); queries, LLM calls, latency

Ablations

Dependency-aware vs. marginal sampling; group vs. single-feature; intervention vs. SHAP-sign verdicts; selective vs. full repair, with and without re-verification; reference, ε, extractor errors.

Pilot and Go/No-Go Gate

Two-week pilot (19 Oct–1 Nov 2026) on about 200 denied Home Credit applicants with three LLMs, the verifier, and the FAX-style baseline (B4); go/no-go meeting proposed for Mon 2 Nov 2026. Thresholds are pre-registered and illustrative.

- **G1, problem exists:** at least 10% of explanations have a direction, prominence, or omission error (E2, E3, E4).
- **G2, gap exists:** the FAX-style baseline leaves at least 30% of the verifier's errors undetected or unrepaired (threshold proposed by the team).
- **G3, verifier is sound:** direction verdicts match the analytic LR reference: close to 100% with the marginal sampler (implementation check), at least 95% with the conditional kNN sampler.

If the gate fails

If general verification already handles the errors, or LLM explanations are rarely wrong, reconsider the direction rather than forcing it.

Section 6

Expected Contributions

Expected Contributions

1. A regulation-informed error taxonomy and benchmark for credit denial explanations, with claim-level labels from model interventions.
2. A dependency-aware, intervention-based claim verifier, with analysis of where SHAP-sign checks disagree.
3. A selective claim-level repair procedure, compared with FAX-style filtering, RARR-style revision, and full regeneration.
4. An empirical study across LLMs, prompts, and datasets of which errors occur and which strategies catch them.

Current Status and Next Steps

Initial idea from a targeted literature search; scaffold code, German Credit run and a pilot pre-registration are in place (15 October 2026). Next: full-text review of FAX and the LLM-credit studies (Week 0), the two-week pilot (19 Oct–1 Nov) and the go/no-go meeting on 2 Nov, then refine or reconsider the direction.

Risks and Mitigations

Risk	Mitigation
Overlap with FAX	Re-implement it as B4; build on credit error types, correlated-feature interventions, preservation-aware repair; test the gap in the pilot
Reference validity	State the definition openly; report sensitivity; validate on LR, semi-synthetic data, expert labels
Implausible interventions	Group-level changes, derived-feature recomputation, conditional sampling; ablate against marginals
LLM variance and cost	Several samples per applicant, fixed decoding, at least two open-weight models; fewer samples and cached outputs if over budget
Regulatory overreach	Call the taxonomy regulation-informed; make no legal-compliance claim
Data access and licensing	Start with Home Credit and German Credit; confirm terms of other sources
Delay in Weeks 3–5	Simplify the schema, keep only the kNN sampler, postpone the generative sampler
No expert annotators	Ask the supervisor in Week 0; fallback: finance students with a written guide
Application paper at an NLP venue	Stress benchmark, taxonomy, analysis of strategies; target a finance-NLP workshop first

Timeline

Weeks	Dates	Work	Output / decision
0	10–18 Oct 2026	Supervisor meeting, full-text reading, data and licences, environment	Related-work table, working environment
1–2	19 Oct–1 Nov	Data, reason groups, generators, pilot with FAX-style baseline	Go/no-go meeting, Mon 2 Nov
3–5	2–22 Nov	Extractor, taxonomy, verifier, LR and semi-synthetic validation	Validated verifier
6–8	23 Nov–13 Dec	Selective repair, remaining baselines	End-to-end system
9–11	14 Dec–3 Jan 2027	Full experiments, ablations, expert annotation	Main results
12–13	4–17 Jan 2027	Writing, figures, full-text related-work check	Submission draft

- Target: a finance-NLP workshop (e.g., FinNLP); ACL 2027 via ARR as a stretch goal (dates to be confirmed in Week 0).
- Weeks 9–11 overlap with Christmas and New Year: schedule long generation and intervention jobs then.
- Fallback scope if there is no end-to-end run by Week 8: drop Lending Club, Freddie Mac, random forest and E6.

Thank you for your attention!